1009 U2 normal distribution

1. Scientists estimate that the distribution of the life span of the Galápagos Islands giant tortoise is approximately normal with mean 100 years and standard deviation 15 years. Based on the estimate, which of the following is closest to the age of a Galápagos Islands giant tortoise at the 90th percentile of the distribution?

(A) 115 years

(B) 120 years ✓

(C) 125 years

(D) 130 years

Answer B

Correct. The value of approximately 119.22, found using technology, is the value that has 90 percent of the area to the left of it in the normal distribution with mean 100 and standard deviation 15. Of the values listed, 120 is the tortoise age that is closest to 119.22.

2. The height requirement for a certain ride at an amusement park is 48 inches or taller. The distribution of height of six-year-old children is approximately normal with mean 45.2 inches and standard deviation 1.75 inches.

Which of the following is closest to the probability that a randomly selected six-year-old child will not meet the height requirement for the ride?

(A) 0.0401

(B) 0.0548

(C) 0.9452 ✓

(D) 0.9599

Answer C

Correct. Let X represent the height of a six-year-old child. The probability that a randomly selected six-year-old child will not meet the height requirement for the ride is $\mathrm{P}(\mathrm{X} < 48) = \mathrm{P}\left(\mathrm{Z} < \frac{48-45.2}{1.75}\right) = \mathrm{P}\left(\mathrm{Z} < 1.6\right) \approx 0.9452$.

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3. The distribution of the number of transactions per day at a certain automated teller machine (ATM) is approximately normal with a mean of 80 transactions and a standard deviation of 10 transactions. Which of the following represents the parameters of the distribution?

(A) $\overline{x} = 80; \quad s = 10$

(B) $\overline{x} = 80; \quad \sigma = 10$

(C) $\mu = 80; \quad \sigma = 10$ ✓

(D) $\mu = 80; \quad s = 10$

Answer C

Correct. The values provided in the question represent known parameter values for the distribution, and the correct symbols for these values are μ (for the population mean) and σ (for the population standard deviation).

4. In the Dominican Republic in August, the distribution of daily high temperature is approximately normal with mean 86 degrees Fahrenheit ($^\circ\text{F}$). Approximately 95% of all daily high temperatures are between 83°F and 89°F. What is the standard deviation of the distribution?

(A) 1°F

(B) 1.5°F ✓

(C) 2°F

(D) 3°F

Answer B

Correct. The empirical rule states that approximately 95% of the observations in a data set that is approximated by a normal curve will lie within 2 standard deviations of the mean. Since 95% of all daily high temperatures are between 83°F and 89°F, the interval from 83°F to 89°F has a width of 4 standard deviations (2 above the mean and 2 below the mean). So 1 standard deviation is $\frac{89-83}{4}\,^\circ\text{F} = 1.5^\circ\text{F}$.

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5. The life span of a battery is the amount of time the battery will last. The distribution of life span for a certain type of battery is approximately normal with mean 2.5 hours and standard deviation 0.25 hour. Suppose one battery will be selected at random. Which of the following is closest to the probability that the selected battery will have a life span of at most 2.1 hours?

(A) 0.055 ✓

(B) 0.110

(C) 0.445

(D) 0.945

Answer A

Correct. If x represents the number of hours the battery will last, then $P(x < 2.1) = P\left(\frac{x-\mu}{\sigma} < \frac{2.1-2.5}{0.25}\right) = P(z < -1.6) \approx 0.055$. Of the values listed, 0.055 is closest to the probability that the selected battery will have a life span of at most 2.1 hours.

6. The distribution of body temperature for healthy adults is approximately normal with mean 98.2°F and standard deviation 0.7°F. Based on the description of the distribution, which of the following statements CANNOT be supported?

(A) More than 95% of healthy adults have a body temperature greater than 96.8°F.

(B) The first quartile of the distribution of healthy adults' body temperature is less than 97.5°F. ✓

(C) Approximately 50% of healthy adults have a body temperature greater than 98.2°F.

(D) Approximately 95% of healthy adults have a body temperature between 96.8°F and 99.6°F.

Answer B

Correct. The first quartile of an approximately normal distribution with mean of 98.2°F and standard deviation 0.7°F is 97.728°F, which is greater than, not less than, 97.5°F. The statement that the first quartile of the distribution of healthy adults' body temperature is less than 97.5°F CANNOT be supported by the description of the distribution.

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7. The distribution of sale price for a certain model of car is approximately normal with mean $\$22{,}500$ and standard deviation $\$600$.

Based on the distribution, which statement must be true?

(A) Approximately 99% of the cars have a sale price less than $\$23{,}700$.

(B) Approximately 84% of the cars have a sale price greater than $\$21{,}900$. ✓

(C) Approximately 95% of the cars have a sale price between $\$21{,}900$ and $\$23{,}100$.

(D) The minimum sale price is $\$20{,}700$.

Answer B

Correct. According to the empirical rule, approximately 68% of the sale prices are within 1 standard deviation of the mean, so $100\% - 68\% = 32\%$ of the sale prices will fall outside of the interval $\$21{,}900$ to $\$23{,}100$. Since the normal distribution is symmetric around the mean, 16% of the sale prices will be less than $\$21{,}900$. Therefore, approximately 84% of the cars will have a sale price greater than $\$21{,}900$.

8. A family has two cats named Gordo and Flaco. Gordo weighs 15 pounds and Flaco weighs 8 pounds. A cat's weight is classified as unhealthy if the weight is located in the top 5% or bottom 5% of all cat weights. The distribution of cat weights is approximately normal with mean 9.5 pounds and standard deviation 1.5 pounds. Which of the following is the best description of Gordo's and Flaco's weights?

(A) Neither Gordo's nor Flaco's weight is classified as unhealthy.

(B) Gordo's weight is classified as unhealthy but Flaco's weight is not. ✓

(C) Flaco's weight is classified as unhealthy but Gordo's weight is not.

(D) Flaco's weight and Gordo's weight are both classified as unhealthy.

Answer B

Correct. Because Gordo's weight is over 3 standard deviations above the mean, his weight is in the top 5% of all weights and is classified as unhealthy. Flaco's weight is less than 2 standard deviations below the mean, so his weight is not in the bottom 5% of all weights and is not classified as unhealthy.

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9. At a small coffee shop, the distribution of the number of seconds it takes for a cashier to process an order is approximately normal with mean 276 seconds and standard deviation 38 seconds. Which of the following is closest to the proportion of orders that are processed in less than 240 seconds?

(A) 0.17 ✓

(B) 0.36

(C) 0.83

(D) 0.95

Answer A

Correct. The correct z-score is $z = \frac{240-276}{38} = -0.95$. The area under the standard normal curve to the left of that value is approximately 0.17.

10. At a college the scores on the chemistry final exam are approximately normally distributed, with a mean of 75 and a standard deviation of 12. The scores on the calculus final are also approximately normally distributed, with a mean of 80 and a standard deviation of 8. A student scored 81 on the chemistry final and 84 on the calculus final. Relative to the students in each respective class, in which subject did this student do better?

(A) The student did better in chemistry.

(B) The student did better in calculus.

(C) The student did equally well in each course. ✓

(D) There is not enough information for comparison, because the number of students in each class is not known.

11. A sleep time of 15.9 hours per day for a newborn baby is at the 10th percentile of the distribution of sleep times for all newborn babies. Assuming the distribution is normal with standard deviation 0.5 hour, approximately what is the mean sleep time, in hours per day, for newborn babies?

(A) 15.3

(B) 16.3

(C) 16.5 ✓

(D) 16.7

12. The weight of adult male grizzly bears living in the wild in the continental United States is approximately normally distributed with a mean of 500 pounds and a standard deviation of 50 pounds. The weight of adult female grizzly bears is approximately normally distributed with a mean of 300 pounds and a standard deviation of 40 pounds. Approximately, what would be the weight of a female grizzly bear with the same standardized score (z-score) as a male grizzly bear with a weight of 530 pounds?

(A) 276 pounds

(B) 324 pounds ✓

(C) 330 pounds

(D) 340 pounds

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13. The distribution of the amount of a customer's purchase at a convenience store is approximately normal, with mean $15.50 and standard deviation 1.72. Which of the following is closest to the proportion of customer purchase amounts between $14.00 and 16.00 ?

(A) 0.39

(B) 0.42 ✓

(C) 0.61

(D) 0.81

Answer B

Correct. The z-score associated with a purchase amount of \$14.00 is $z = \frac{14-15.5}{1.72} \approx -0.872$, and the z-score associated with a purchase amount of \$16.00 is $z = \frac{16-15.5}{1.72} \approx 0.291$. The area under the standard normal curve between -0.872 and 0.291 is approximately 0.42, so this is the approximate proportion of customer purchase amounts between $14.00 and 16.00.

14. The distribution of the number of transactions performed at a bank each day is approximately normal with mean 478 transactions and standard deviation 64 transactions. Which of the following is closest to the proportion of daily transactions greater than 350 ?

(A) 0.023

(B) 0.477

(C) 0.954

(D) 0.977 ✓

Answer D

Correct. The value 0.977 is the total proportion of transactions that are greater than 350. The z-score associated with 350 transactions is -2. The proportion of daily transactions greater than 350 can be found using the z-score and technology, such as a calculator, a standard normal table, or computer-generated output.

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15. The distribution of the amount of water used per wash cycle, in gallons, by a particular brand of dishwasher during a wash cycle is approximately normal with mean 4 gallons and standard deviation 0.2 gallon. Which of the following is closest to the probability that more than 3.75 gallons are used for any wash cycle?

(A) 0.11

(B) 0.34

(C) 0.79

(D) 0.89 ✓

Answer D

Correct. If x represents the number of gallons of water used for a wash cycle, then $P(x > 3.75) = P\left(\frac{x-\mu}{\sigma} > \frac{3.75-4}{0.2}\right) = P(z > -1.25) \approx 0.89$.

16. Data on homes recently sold in a certain town included the area of the home, reported in square feet. The table below shows summary statistics of the reported areas, in square feet.

Mean	Minimum	Q1	Median	Q3	Maximum	Standard Deviation
1,754.14	1,656	1,704	1,758	1,806	1,843	61.0723

An auditor determined that an error was made in the reported areas and that all of the areas should have been 100 square feet greater than what was reported. The areas were corrected and new summary statistics were reported.

What are the interquartile range (IQR) and the standard deviation of the corrected areas?

(A) IQR 102, standard deviation 61.0723 ✓

(B) IQR 102, standard deviation 161.0723

(C) IQR 202, standard deviation 61.0723

(D) IQR 202, standard deviation 161.0723

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17. The mean and standard deviation of the sample data collected on continuous variable x are -0.25 and 0.03, respectively. The following table shows the relative frequencies of the data in the given intervals.

Interval	Relative Frequency
$-0.34 \leq x < -0.31$	0.02
$-0.31 \leq x < -0.28$	0.15
$-0.28 \leq x < -0.25$	0.33
$-0.25 \leq x < -0.22$	0.36
$-0.22 \leq x < -0.19$	0.11
$-0.19 \leq x < -0.16$	0.03

Based on the table, do the data support the use of a normal model to approximate population characteristics?

(A) Yes, because the sum of the relative frequencies is 1.00.

(B) Yes, because the distribution of relative frequencies is very close to the empirical rule for normal models. ✓

(C) No, because the distribution of relative frequencies is very far from the empirical rule for normal models.

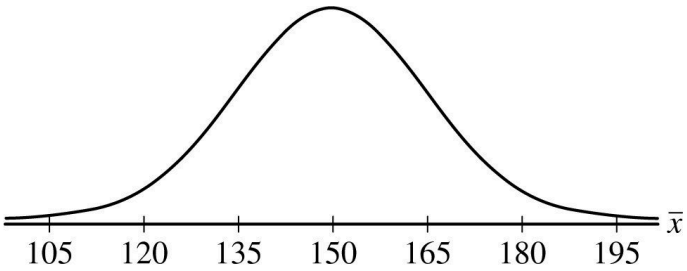
(D) No, because the sample size and the population parameters are not known.

Answer B

Correct. Based on the table, 69% of the data falls within 1 standard deviation of the mean, 95% falls within 2 standard deviations of the mean, and 100% falls within 3 standard deviations of the mean. This is very close to the empirical rule of 68% of the data within 1 standard deviation of the mean, 95% within 2 standard deviations of the mean, and 99.7% within 3 standard deviations of the mean.

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18. The normal curve shown represents the sampling distribution of a sample mean for sample size $n = 25$, selected at random from a population with standard deviation σ_x.



Sampling Distribution of $\bar{x}$

Which of the following is the best estimate of the standard deviation of the population, σ_x?

(A) 3

(B) 15

(C) 30

(D) 75 ✓

19. A researcher is studying a group of field mice. The distribution of the weight of field mice is approximately normal with mean 25 grams and standard deviation 4 grams. Which of the following is closest to the proportion of field mice with a weight greater than 33 grams?

(A) 0.023 ✓

(B) 0.046

(C) 0.954

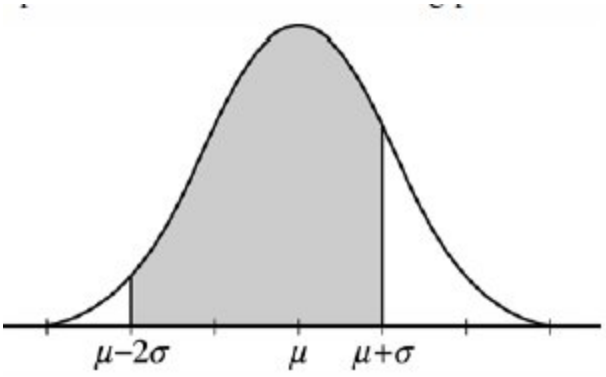
(D) 0.977

Answer A

Correct. The z-score associated with a weight of 33 grams is 2. The area under the curve to the right of 2 in a standard normal distribution is approximately 0.023.

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20. A certain type of remote-control car has a fully charged battery at the time of purchase. The distribution of running times of cars of this type, before they require recharging of the battery for the first time after its period of initial use, is approximately normal with a mean of 80 minutes and a standard deviation of 2.5 minutes. The shaded area in the figure below represents which of the following probabilities?



(A) The probability that the running time of a randomly selected car of this type, before it requires recharging of the battery for the first time after its period of initial use, is between 75 minutes and 82.5 minutes. ✓

(B) The probability that the running time of a randomly selected car of this type, before it requires recharging of the battery for the first time after its period of initial use, is between 75 minutes and 85 minutes.

(C) The probability that the running time of a randomly selected car of this type, before it requires recharging of the battery for the first time after its period of initial use, is between 77.5 minutes and 82.5 minutes.

(D) The probability that the running time of a randomly selected car of this type, before it requires recharging of the battery for the first time after its period of initial use, is between 77.5 minutes and 85 minutes.